



# Quality Assurance Report

| CUSTOMER(고객명) | DESCRIPTION(품명) | LOT NO(로트번호)      | ISSUED DATE(발행일자) |
|---------------|-----------------|-------------------|-------------------|
| 세종에이앤에스       | 0.15*141        | 20250616 -01 ~ 02 | 2025-06-16        |
| 규격(재질)        |                 | LOT SIZE(수량)      |                   |
| 0.15*141 Cu   | Ni 1.0~2.0μm    | 572.4 Kg          |                   |

## I. VISUAL INSPECTION (Sampling Plan : n=100mm/Reel, 5Reel/Shipping or n=50ea/S)

| NO | INSPECTION ITEMS  | Ac/Re | RESULT | NO | INSPECTION        |
|----|-------------------|-------|--------|----|-------------------|
| 1  | DISCOLORATION(변색) | 0/1   | OK     | 6  | NO PLATING(미도금)   |
| 2  | STAIN(얼룩)         | 0/1   | OK     | 7  | PEELING(도금 벗겨짐)   |
| 3  | RUST(부식)          | 0/1   | OK     | 8  | LUMP(도금물침)        |
| 4  | CONTAMINATION(오염) | 0/1   | OK     | 9  | CAMBER(휨)         |
| 5  | BENT(변형)          | 0/1   | OK     | 10 | PLATING BURR (도금) |

## II. DIMENSION INSPECTION (Sampling Plan : n=100mm/Reel, 5Reel/Shipping or n=50ea/S)

| NO | 항목/구분 | SPEC  |       |       | 검사주기     | 우리하이피   |    |
|----|-------|-------|-------|-------|----------|---------|----|
|    |       | 규격    | 하한(-) | 상한(+) |          | 1차      | 2차 |
| 1  | 전폭    | 141.0 | -0.1  |       | Reel/Lot | 140.999 |    |
| 2  |       |       |       |       |          |         |    |
| 3  |       |       |       |       |          |         |    |
| 4  |       |       |       |       |          |         |    |
| 5  |       |       |       |       |          |         |    |
| 6  |       |       |       |       |          |         |    |
| 7  |       |       |       |       |          |         |    |
| 8  |       |       |       |       |          |         |    |
| 9  |       |       |       |       |          |         |    |
| 10 |       |       |       |       |          |         |    |

6/16  
0.15 x 141  
① 286.2  
② 286.2  
총 472.4kg

## III. PLATING THICKNESS(도금두께) (Sampling Plan : 1y)

| NO | SPECIFICATION            | Measurement Data | RESULT |
|----|--------------------------|------------------|--------|
| 1  | X-Ray 측정<br>Ni 1.0~2.0μm | 첨부 : 도금성적서       | OK     |
|    |                          | 첨부 : 도금성적서       |        |

## IV. FUNCTIONAL TEST (S)

| NO | INSPECTION           | RESULT |
|----|----------------------|--------|
| 1  | ADHESION<br>(밀착성 시험) | OK     |

|      |  |     |     |     |      |  |
|------|--|-----|-----|-----|------|--|
| 제품도면 |  | 하이피 | 김영란 | 조승현 | 합판정  |  |
|      |  | 고객  | 작성  | 승인  | 종합판정 |  |

업체명 : 세종에이앤에스

품명 : 0.15x141      초물      LOT NO : 250616-01      DATE : 25.06.16.

| No    | $\mu\text{m}$<br>Ni | Ni tolerance |       |       |
|-------|---------------------|--------------|-------|-------|
|       |                     | 1.206        | 1.455 | 1.703 |
| 1     | 1.452               |              |       |       |
| 2     | 1.412               |              |       |       |
| 3     | 1.703               |              |       |       |
| 4     | 1.329               |              |       |       |
| 5     | 1.377               |              |       |       |
| Mean  | 1.455               |              |       |       |
| S.Dev | 0.146               |              |       |       |

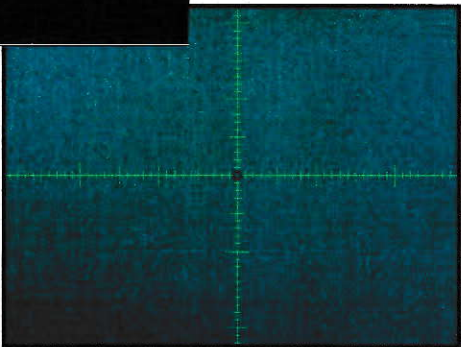
FUNC: MEASURE  
COL : 0.100 mm

Ni  
Cu

SEC : 15 Sec

표준편차      0.146  
변동계수      10.040

우리하이피



업체명 : 세종에이앤에스

품명 : 0.15x141

종물

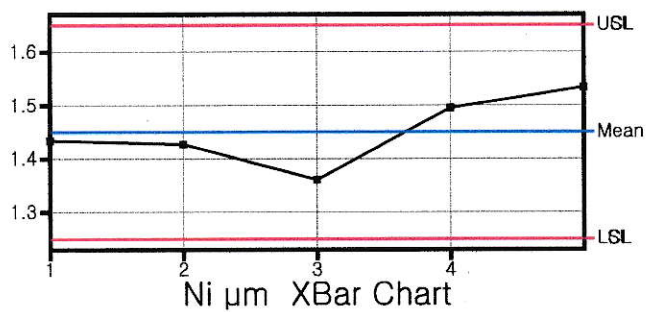
LOT NO : 250616-01

DATE : 25.06.16.

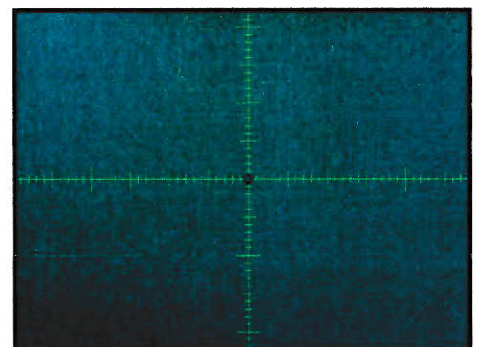
| No    | $\mu\text{m}$<br>Ni | Ni tolerance      |
|-------|---------------------|-------------------|
| 1     | 1.434               | 1.361 1.450 1.539 |
| 2     | 1.427               |                   |
| 3     | 1.361               |                   |
| 4     | 1.495               |                   |
| 5     | 1.534               |                   |
| Mean  | 1.450               |                   |
| S.Dev | 0.067               |                   |

FUNC: MEASURE  
COL : 0.100 mm

$\frac{\text{Ni}}{\text{Cu}}$   
SEC : 15 Sec



도금원소 : Ni//Cu  
측정값 개수 5  
최대값 1.534  
최소값 1.361  
범위 0.173  
평균값 1.450  
표준편차 0.067  
변동계수 4.598



우리하이피